

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	31 January 3035
Team ID	
Project Name	Credit Card Approval Prediction System
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web interface for applicants and admin dashboard	HTML, CSS, JavaScript, React.js
2.	Application Logic-1	Handles user registration, login, and form submission	Python (Flask Framework)
3.	Application Logic-2	Processes applicant data and interacts with ML model	Flask REST API
4.	Application Logic-3	Generates prediction results and sends responses	Scikit-learn / TensorFlow
5.	Database	Stores applicant and prediction data	MySQL
6.	Cloud Database	Cloud-hosted database for scalability	IBM DB2 / AWS RDS
7.	File Storage	Stores uploaded documents securely	IBM Cloud Object Storage / AWS S3
8.	External API-1	Email confirmation and OTP verification	Twilio / SendGrid API
9.	External API-2	Identity verification	Aadhar API
10.	Machine Learning Model	Predicts credit card approval based on applicant data	Logistic Regression / Random Forest
11.	Infrastructure (Server / Cloud)	Deployment environment	IBM Cloud / AWS EC2 / Kubernetes

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frameworks used for development and ML	Flask, Scikit-learn, TensorFlow
2.	Security Implementations	Encryption, authentication, and access control	SHA-256, HTTPS, JWT Tokens

S.No	Characteristics	Description	Technology
3.	Scalable Architecture	Supports multiple users and data growth	3-Tier Architecture (Frontend, Backend, Database)
4.	Availability	Ensures uptime and load balancing	AWS Load Balancer / IBM Cloud Auto-Scaling
5.	Performance	Optimized for fast predictions and response time	Caching, CDN, optimized ML inference